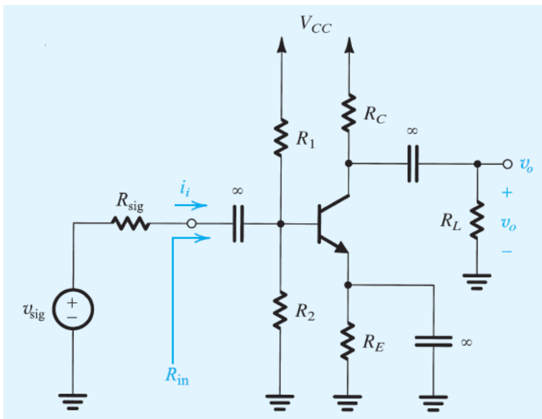
ECE 3355: Electronics

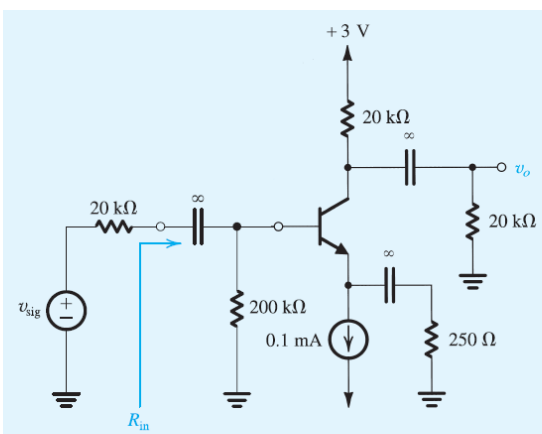
Homework 10

Topic: BJTs Small Signal Analysis

1. For the common-emitter amplifier shown below, let V_{CC}=15 V, R_1=27 kΩ, R_2=15 kΩ, R_E=2.4 kΩ, and R_C=3.9 kΩ. The transistor has β=100. Calculate the dc bias current I_C. If the amplifier operates between a source for which R_{sig}=2 kΩ and a load of 2 kΩ, replace the transistor with its hybrid-π model, and find the values of R_{in}, and the overall voltage gain v_o/v_{sig}.



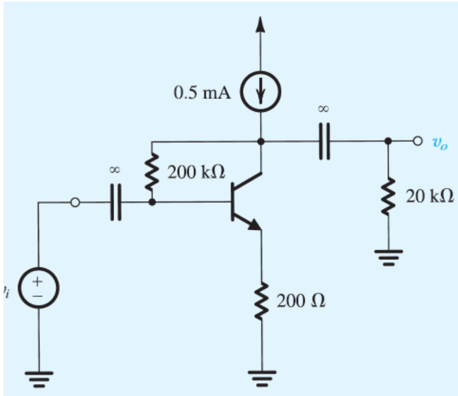
2. In the circuit below, the BJT is biased with a constant-current source, and vsig is a small sine-wave signal. Find R_{in} and the gain v_o/v_{sig}. Assume β=100. If the amplifier of the signal v_{be} is to be limited to 5 mV, what is the largest signal at the input? What is the corresponding signal at the output?



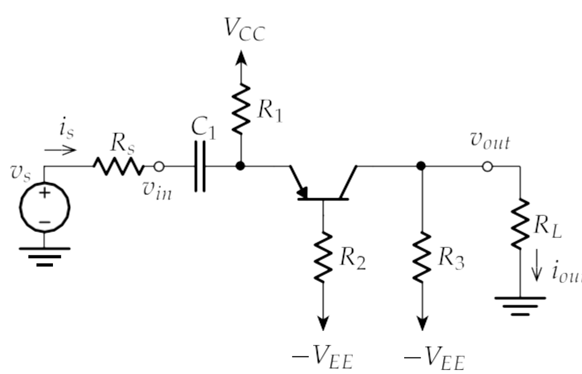
3. The BJT in the circuit below has β=100.

   (a) Find the dc collector current and the dc voltage at the collector.

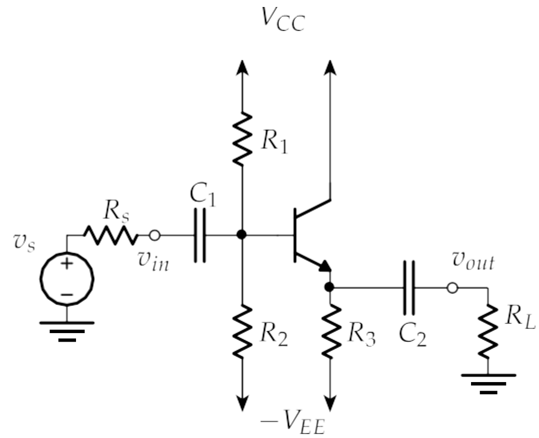
   (b) Replacing the transistor by its π model, draw the small-signal equivalent circuit of the amplifier. Analyze the resulting circuit to determine the voltage gain v_o/v_i.



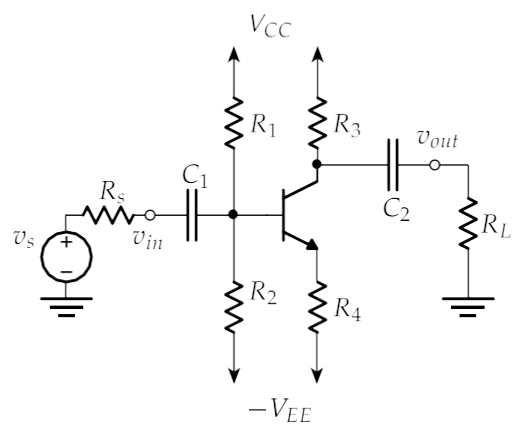
4. For the following circuit, use R_s = 100 Ω, C_1 = 10 µF, R_1 = 1 KΩ, R_2 = 220 KΩ, R_3 = 1.5 kΩ, R_L = 500 Ω, V_{CC} = +15 V, -V_{EE} = -15 V, and β = 100.
   a. Confirm that the circuit is in the active mode
   b. Draw the small signal model of the circuit
   c. Find the amplifier's ac gain (v_{out}/v_{in} with source and load removed)
   d. Find the small signal input resistance (with the load attached)
   e. Find the small signal output resistance (with the source attached)
   f. Sketch the simplified model for the amplifier
   g. Find the voltage gain of the circuit (with the source and load attached)
   h. Find the current gain of the circuit (with the source and load attached)



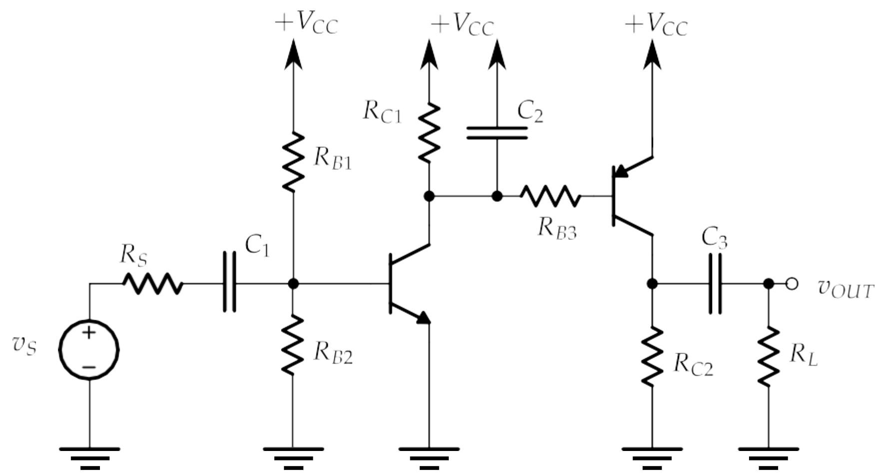
5. For the following circuit, use R_s = 500 Ω, C_1 = 0.1 µF, C_2 = 1 µF, R_1 = R_2 = 50 KΩ, R_3 = 1 KΩ, and R_L = 100 Ω, V_{CC} = +12 V, -V_{EE} = -12 V, and β = 100.
   a. Confirm that the circuit is in the active mode
   b. Draw the small signal model of the circuit
   c. Find the amplifier's ac gain (v_{out}/v_{in} with source and load removed)
   d. Find the small signal input resistance (with the load attached)
   e. Find the small signal output resistance (with the source attached)
   f. Sketch the simplified model for the amplifier
   g. Find the gain of the circuit
   h. Sketch the Bode magnitude plot



6. For the following circuit, use R_s = 1 KΩ, C_1 = C_2 = 1μF, R_1 = 4.7 KΩ, R_2 = 2.2 KΩ, R_3 = 1 KΩ, R_4 = 1 KΩ, R_L = 500 Ω, V_{CC} = +15 V, - V_{EE} = -15 V, and β = 100.
   a. Confirm that the circuit is in the active mode
   b. Draw the small signal model of the circuit
   c. Find the amplifier's ac gain (v_{out}/ v_{in} with source and load removed)
   d. Find the small signal input resistance (with the load attached)
   e. Find the small signal output resistance (with the source attached)
   f. Find the gain of the circuit
   g. Using the simplified amplifier model, sketch the Bode magnitude plot



7. Solve the following problem.



(a) Complete a DC analysis to find the values of the small signal model and ensure that the transistor in an active mode using V_{CC} = 20 V, R_s = 10 Ω, R_{B1} = R_{B2} = R_{B3} = 100 KΩ, R_{C1} = 540 Ω, R_{C2} = 1075 Ω, R_L = 1000 Ω, C_1 = 75 μF, C_2 = 186 nF, C_3 = 48 μF, and β = 100. Use the DC analysis to compute the parameters for the small signal analysis.

(b) Write an expression (no values!) for the *midband* gain, v_{out}/v_{in}, with the source and load removed. Note that v_{in} is the node between R_s and C_1.

(c) Find the input and output resistances.

(d) Estimate the RC time constants for each of the stages of the amplifier.

(e) Sketch the Bode magnitude plot for the circuit.